

Let's solve this step by step 📌

Given:

The point with respect to the moving coordinate frame is

$$P_m = \begin{bmatrix} 0.5 \\ 0.8 \\ 1.3 \\ 1 \end{bmatrix}$$

The moving frame is rotated by 90° about the z-axis of the fixed frame.

Step 1: Write the rotation matrix for 90° about the z-axis

Rotation about z-axis (counterclockwise) by $+90^\circ$:

$$R_z(90^\circ) = \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 2: Find coordinates of P in the fixed frame

The coordinates of the same point with respect to the fixed frame are:

$$P_f = R_z(90^\circ) \times P_m$$

Ignoring the homogeneous coordinate (last "1"):

$$\begin{bmatrix} x_f \\ y_f \\ z_f \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.8 \\ 1.3 \end{bmatrix} = \begin{bmatrix} -(0.8) \\ 0.5 \\ 1.3 \end{bmatrix} = \begin{bmatrix} -0.8 \\ 0.5 \\ 1.3 \end{bmatrix}$$

✅ **Final Answer:**

$$P_f = \begin{bmatrix} -0.8 \\ 0.5 \\ 1.3 \\ 1 \end{bmatrix}$$

Interpretation:

When the moving frame is rotated 90° counterclockwise about the z-axis, the x-y coordinates of the point are rotated accordingly — the x-coordinate becomes negative y, and the y-coordinate becomes positive x.

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Problem 2.2

Question:

Determine the rotation matrix for a rotation of 45° about y-axis, followed by a rotation of 120° about z-axis, and a final rotation of 90° about x-axis.

Step 1: Write the standard rotation matrices.

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 2: Order of rotation

The vector is first rotated about y, then z, then x.

$$R = R_x(90^\circ)R_z(120^\circ)R_y(45^\circ)$$

Step 3: Substitute values

$$R_y(45^\circ) = \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ -0.7071 & 0 & 0.7071 \end{bmatrix}$$

$$R_z(120^\circ) = \begin{bmatrix} -0.5 & -0.8660 & 0 \\ 0.8660 & -0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_x(90^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

Step 4: Multiply step-by-step

First $R_z(120^\circ)R_y(45^\circ)$:

$$R_{zy} = \begin{bmatrix} -0.3536 & -0.8660 & -0.6124 \\ 0.6124 & -0.5 & 0.6124 \\ -0.7071 & 0 & 0.7071 \end{bmatrix}$$

Now multiply with $R_x(90^\circ)$:

$$R = R_x(90^\circ)R_{zy} = \begin{bmatrix} -0.3536 & -0.8660 & -0.6124 \\ 0.7071 & 0 & 0.7071 \\ -0.6124 & 0.5 & -0.6124 \end{bmatrix}$$

✅ Final Rotation Matrix:

$$R = \begin{bmatrix} -0.354 & -0.866 & -0.612 \\ 0.707 & 0 & 0.707 \\ -0.612 & 0.5 & -0.612 \end{bmatrix}$$

Problem 2.3

Given:

$$P = 3i - 2j + 5k = [3, -2, 5]^T$$

- Rotate by 90° about x-axis
 - Then by 90° about z-axis
 - Then translate by $-3i + 2j - 5k = [-3, 2, -5]^T$
-

Step 1: Apply rotation about x-axis (90°):

$$R_x(90^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P_1 = R_x P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \\ -2 \end{bmatrix}$$

Step 2: Rotate P_1 about z-axis (90°):

$$R_z(90^\circ) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P_2 = R_z P_1 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -5 \\ -2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$$

Step 3: Translation

$$P_{final} = P_2 + [-3, 2, -5]^T = [5 - 3, 3 + 2, -2 - 5]^T = [2, 5, -7]^T$$

✓ Final position of vector P:

$$P' = 2i + 5j - 7k$$



Problem 2.4

Given:

$$G = [3, 0, -1, 1]^T$$

We'll treat homogeneous coordinates for clarity.

(i) Rotate by π about z-axis, then translate by 3 along y-axis

$$R_z(\pi) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_y(3) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_z(\pi) = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$G_1 = T_y(3)R_z(\pi)G$$

Compute step-by-step:

After rotation:

$$R_z(\pi)G = \begin{bmatrix} -3 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

Then translation:

$$G_1 = \begin{bmatrix} -3 \\ 3 \\ -1 \\ 1 \end{bmatrix}$$

(ii) Translate first, then rotate

Translate by 3 along y-axis first:

$$G' = [3, 3, -1, 1]^T$$

Then rotate by π about z-axis:

$$G_2 = R_z(\pi)G' = \begin{bmatrix} -3 \\ -3 \\ -1 \\ 1 \end{bmatrix}$$

✓ Final positions:

- Case (i): $G_1 = [-3, 3, -1]^T$
- Case (ii): $G_2 = [-3, -3, -1]^T$

Nice — I'll solve the exercises 2.5 → 2.17 shown in your image and explain what I can and where more info is needed. I'll give clear final answers and show the key steps. (Problem 2.18 in the image is cut off so I can't finish that one without the missing text.)

2.5

Problem. A moving frame is rotated about the x -axis of the fixed frame by $\pi/6$. The coordinates of a point Q in the fixed frame are $Q_f = [2 \ 0 \ 3]^T$. What are the coordinates of Q with respect to the moving frame?

Idea. If the moving frame is obtained from the fixed frame by rotation $R_x(\pi/6)$, then the coordinates of the same physical point in the moving frame are

$$Q_m = R_x(\pi/6)^T Q_f$$

(because we want components in the rotated frame).

$$R_x(\pi/6) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \frac{\pi}{6} & -\sin \frac{\pi}{6} \\ 0 & \sin \frac{\pi}{6} & \cos \frac{\pi}{6} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}.$$

So

$$Q_m = R_x(\pi/6)^T Q_f = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ \frac{1}{2} \cdot 3 = 1.5 \\ -\frac{\sqrt{3}}{2} \cdot 3 \approx 2.598076 \end{bmatrix}.$$

Answer: $Q_m = [2, 1.5, 2.598076]^T$ (exact: $[2, 3/2, (3\sqrt{3})/2]^T$).

2.7

Problem. The end-effector is rotated about fixed axes starting with a yaw of $-\pi/2$ (about z), followed by a pitch of $-\pi/2$ (about y). What is the resulting rotation matrix?

Interpretation & solution. “Yaw then pitch” about fixed axes means

$$R = R_y(-\frac{\pi}{2}) R_z(-\frac{\pi}{2}).$$

Compute matrices:

$$R_z(-\frac{\pi}{2}) = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_y(-\frac{\pi}{2}) = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Multiply $R = R_y(-\pi/2)R_z(-\pi/2)$:

$$R = \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Answer: $R = \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$

2.8

Problem. A vector in frame $\{b\}$ is ${}^bC = [2 \ 4 \ -5]^T$. Frame $\{b\}$ is rotated by $-\pi/4$ about the x -axis of frame $\{a\}$. Determine aC .

Solution. Rotation about x by $\theta = -\pi/4$:

$$R_x(-\frac{\pi}{4}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(-45^\circ) & -\sin(-45^\circ) \\ 0 & \sin(-45^\circ) & \cos(-45^\circ) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}.$$

(Here we use the convention ${}^aC = R_{a \leftarrow b} {}^bC$ where $R_{a \leftarrow b} = R_x(-\pi/4)$.)

Multiply:

$${}^aC = R_x(-\frac{\pi}{4}) \begin{bmatrix} 2 \\ 4 \\ -5 \end{bmatrix} = \begin{bmatrix} 2 \\ \frac{\sqrt{2}}{2}(4) + \frac{\sqrt{2}}{2}(-5) \\ -\frac{\sqrt{2}}{2}(4) + \frac{\sqrt{2}}{2}(-5) \end{bmatrix} \approx \begin{bmatrix} 2 \\ -0.7071 \\ -6.36396 \end{bmatrix}.$$

(Exact: second component $= \frac{\sqrt{2}}{2}(4 - 5) = -\frac{\sqrt{2}}{2}$, third $= -\frac{\sqrt{2}}{2}(4 + 5) = -\frac{9\sqrt{2}}{2}$.)

Answer: ${}^aC \approx [2, -0.7071, -6.36396]^T.$

2.9

Problem. If bC from 2.8 is also translated by 4 units in the $-y$ direction (in addition to the rotation), determine aC .

Solution. Translation (expressed in frame a) $d = [0, -4, 0]^T$. After rotation and translation:

$${}^aC = R_x(-\frac{\pi}{4}) {}^bC + d.$$

Using numeric result from 2.8:

$${}^aC \approx [2, -0.7071, -6.36396]^T + [0, -4, 0]^T = [2, -4.7071, -6.36396]^T.$$

Answer: ${}^aC \approx [2, -4.7071, -6.36396]^T$.

2.10

Problem. The end-effector holds a tool whose tip in end-effector coordinates is $P = [0 \ 0 \ 1.2]^T$. Find tool-tip coordinates with respect to a fixed base frame, if the end-effector pose (homogeneous transform) is given by T_{base}^{ee} (Eq. 2.31 in your book).

Solution (general formula). In homogeneous form,

$$P_{ee}^h = \begin{bmatrix} 0 \\ 0 \\ 1.2 \\ 1 \end{bmatrix}, \quad P_{base}^h = T_{base}^{ee} P_{ee}^h.$$

So the tool tip in the base frame is the top 3 elements of $T_{base}^{ee} [0 \ 0 \ 1.2 \ 1]^T$.

Answer: $P_{base} = (T_{base}^{ee} [0, 0, 1.2, 1]^T)_{1:3}$.

(If you give the numeric T_{base}^{ee} I can compute the explicit coordinates.)

2.10

Problem. The end-effector holds a tool whose tip in end-effector coordinates is $P = [0 \ 0 \ 1.2]^T$. Find tool-tip coordinates with respect to a fixed base frame, if the end-effector pose (homogeneous transform) is given by T_{base}^{ee} (Eq. 2.31 in your book).

Solution (general formula). In homogeneous form,

$$P_{ee}^h = \begin{bmatrix} 0 \\ 0 \\ 1.2 \\ 1 \end{bmatrix}, \quad P_{base}^h = T_{base}^{ee} P_{ee}^h.$$

So the tool tip in the base frame is the top 3 elements of $T_{base}^{ee} [0 \ 0 \ 1.2 \ 1]^T$.

Answer: $P_{base} = (T_{base}^{ee} [0, 0, 1.2, 1]^T)_{1:3}$.

(If you give the numeric T_{base}^{ee} I can compute the explicit coordinates.)

2.11

Problem. A point Q is located 8 units along the y -axis of a moving frame. The mobile frame, initially coincident with the fixed frame, is rotated by $\pi/3$ about the fixed z -axis. Determine coordinates of Q in the fixed frame.

Solution. In the moving frame $Q_m = [0, 8, 0]^T$. After rotation about fixed z by $\pi/3$,

$$R_z\left(\frac{\pi}{3}\right) = \begin{bmatrix} \cos 60^\circ & -\sin 60^\circ & 0 \\ \sin 60^\circ & \cos 60^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.8660254 & 0 \\ 0.8660254 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

So

$$Q_{fixed} = R_z\left(\frac{\pi}{3}\right)[0, 8, 0]^T = \begin{bmatrix} -0.8660254 \cdot 8 \\ 0.5 \cdot 8 \\ 0 \end{bmatrix} \approx \begin{bmatrix} -6.9282 \\ 4 \\ 0 \end{bmatrix}.$$

Answer: $Q_{fixed} \approx [-6.9282, 4, 0]^T$.



2.12

Problem. The end-effector is a gripper. The gripper is relocated from initial point $[2 \ 0 \ 4]^T$ to $[4 \ 0 \ 0]^T$. Determine the direction of axis $\mathbf{k}$ and the angle of rotation about this $\mathbf{k}$ -axis.

Comment & analysis. A pure rotation about a single axis through the origin preserves the distance from the origin. Compare norms:

$$\|[2, 0, 4]\| = \sqrt{2^2 + 4^2} = \sqrt{20} \approx 4.4721, \quad \|[4, 0, 0]\| = 4.$$

These norms are different, so **no single rotation about an axis through the origin** can map the first point to the second without additional translation. There are infinitely many combinations of rotation about some axis not passing through the origin (or rotation + translation) that can perform the relocation, but the axis and angle are **not uniquely determined** by these two points alone.

Answer: Not enough information to find a unique axis and angle for a **pure rotation** — the distance changed so a pure rotation about a fixed axis through the origin is impossible. If you mean rotation about an axis that passes through some other point (i.e., a rigid motion combining rotation and translation), please give either the axis location or specify that a translation is allowed; then we can compute axis/angle (there are many possibilities unless you fix extra constraints).

2.13

Problem. Show that a rotation by θ about axis $\mathbf{k}$ can reproduce the fundamental rotations when $\mathbf{k}$ is chosen as the x -, y -, or z -axis.

Solution (short): The axis-angle formula (Rodrigues) gives the rotation matrix for rotation by θ about a unit axis $\mathbf{k} = (k_x, k_y, k_z)$:

$$R(\theta, \mathbf{k}) = I \cos \theta + (1 - \cos \theta) \mathbf{k} \mathbf{k}^T + (\sin \theta) [\mathbf{k}]_{\times}.$$

If $\mathbf{k} = (1, 0, 0)^T$ this reduces to $R_x(\theta)$; if $\mathbf{k} = (0, 1, 0)^T$ it reduces to $R_y(\theta)$; if $\mathbf{k} = (0, 0, 1)^T$ it reduces to $R_z(\theta)$. (Plugging in the three unit axes shows identical forms to the standard fundamental rotation matrices.)

Answer: Done by substitution into Rodrigues' formula.



2.14

Problem. An end-effector is rotated by 60° about an axis whose unit vector is (given in problem) $\hat{k} = [1/\sqrt{2}, 1/\sqrt{2}, 1]^T$ — verify normalization then find the homogeneous transformation matrix representing this rotation.

Clarify & compute. First normalize that vector: call $u = [1/\sqrt{2}, 1/\sqrt{2}, 1]^T$. Its norm is

$$\|u\| = \sqrt{\frac{1}{2} + \frac{1}{2} + 1} = \sqrt{2} = \sqrt{2}.$$

So the unit axis is

$$\hat{k} = \frac{u}{\sqrt{2}} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \approx \begin{bmatrix} 0.5 \\ 0.5 \\ 0.70710678 \end{bmatrix}.$$

Use Rodrigues' formula with $\theta = 60^\circ$ ($\cos=0.5$, $\sin \approx 0.8660254$). Let $u = (u_x, u_y, u_z)$ denote this unit axis. The rotation matrix is

$$R = I \cos \theta + (1 - \cos \theta)uu^T + \sin \theta [u]_\times,$$

$$\text{where } [u]_\times = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix}.$$

Using $u \approx [0.5, 0.5, 0.70710678]^T$ and executing the arithmetic gives (rounded):

$$R \approx \begin{bmatrix} 0.6250 & -0.4874 & 0.6098 \\ 0.7374 & 0.6250 & -0.2562 \\ -0.2562 & 0.6098 & 0.7500 \end{bmatrix}.$$

Make homogeneous transform (pure rotation, no translation):

$$T = \begin{bmatrix} R & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}.$$

Answer: $T = \begin{bmatrix} R & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}$ with R above (numerical values shown). If you want higher precision or exact surd expressions, I can provide them.

2.15

Problem. The end-point of a link is at $P = [2 \ 2 \ 6 \ 1]^T$. The link is rotated by 90° about the x -axis, then by -180° about its own w -axis, and finally by -90° about its own v -axis. Find the resulting homogeneous transform and final location of the end-point.

Comment & analysis. This problem refers to rotations about the link's **own** axes w and v , i.e. axes that have been changed by prior rotations. To produce a *numerical* final point we need:

- a clear definition of the initial orientation of the link axes v, w relative to the fixed frame (which axis corresponds to v and w initially?),
- an unambiguous order (are all rotations applied to the point in a frame-attached manner — intrinsic rotations — which I assume), and
- whether the rotations are about the current frame axes passing through the origin (or about the end point).

Given the problem statement alone, the correct procedure is:

1. Build the 3×3 rotation matrices in the correct intrinsic order (for intrinsic rotations about current axes) by **pre- or post- multiplying** appropriately.
2. Form the final homogeneous transform $T = \begin{bmatrix} R_{final} & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}$ (if no translation).
3. Compute $P_{final} = T P$.

But because the problem uses labels v, w without specifying which current axis they correspond to numerically, I cannot give a unique numeric final point. If you confirm which axes v, w match to (for instance: v = current y -axis, w = current z -axis, etc.) I will compute the product explicitly and give the final location.

Answer: procedure given; request you tell me which axes v and w correspond to (or confirm they mean current y and z , for example), and I'll compute the numeric result.

2.16

Problem. For a rotation of 90° about z , followed by 90° about y , followed by 90° about x , determine an equivalent single rotation axis $\mathbf{k}$ and rotation angle θ .

Solution. Interpret as rotations about fixed axes in that order $R = R_z(90^\circ)R_y(90^\circ)R_x(90^\circ)$ (apply z then y then x):

- Using standard matrices and multiplying (see working below) we obtain

$$R = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

- Find axis: vector $\mathbf{k}$ satisfies $(R - I)\mathbf{k} = 0$. Solve $\rightarrow \mathbf{k}$ proportional to $[1, 0, 1]^T$. Unit axis:

$$\mathbf{k} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

- Angle from trace formula: $\text{tr}(R) = -1 = 1 + 2 \cos \theta \Rightarrow \cos \theta = -1 \Rightarrow \theta = \pi$ (180°).

Answer: Equivalent rotation is a rotation by π (180°) about axis $\mathbf{k} = \frac{1}{\sqrt{2}}[1, 0, 1]^T$.

(Working summary: compute product R , then get eigenvector for eigenvalue 1 and use trace to get θ .)

2.17

Problem. Determine the homogeneous transform T that represents a translation of 1 unit along the x -axis, followed by a rotation of angle α about x -axis, followed by a rotation of θ about the rotated z -axis.

Solution (matrix form). Apply transformations to a point in this order (first translate, then rotate about x , then rotate about current z). Each operation is represented as a 4×4 homogeneous matrix. With:

$$T_x(1) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha & 0 \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

and a rotation about the current z -axis by θ (intrinsic) has matrix in global coordinates $R_z(\theta)$. The total transform (applied to a point column vector) is:

$$T = R_z(\theta) R_x(\alpha) T_x(1).$$

Compute compactly:

$$R_x(\alpha)T_x(1) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & \cos \alpha & -\sin \alpha & 0 \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

so

$$T = R_z(\theta) \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & \cos \alpha & -\sin \alpha & 0 \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Carrying out the multiplication gives the explicit homogeneous matrix

$$T = \begin{bmatrix} \cos \theta & -\sin \theta \cos \alpha & \sin \theta \sin \alpha & \cos \theta \\ \sin \theta & \cos \theta \cos \alpha & -\cos \theta \sin \alpha & \sin \theta \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

(You can verify the top-right translation column equals $R_z(\theta)[1, 0, 0]^T = [\cos \theta, \sin \theta, 0]^T$.)

Answer: $T = R_z(\theta) R_x(\alpha) T_x(1)$ with the explicit 4×4 matrix above.

We are given a sequence of 4 transformations applied to frame $\{B\}$, initially coincident with frame $\{A\}$, and are to compute the final homogeneous transformation matrix ${}^A T_B$.

✓ Transformations Summary

1. Rotation θ about the Z-axis
2. Translation d along the Z-axis
3. Translation a along the X-axis
4. Rotation α about the X-axis

These transformations follow the Denavit–Hartenberg (D-H) convention and the resulting transformation is:

$${}^A T_B = R_z(\theta) \cdot T_z(d) \cdot T_x(a) \cdot R_x(\alpha)$$

📅 Step-by-Step Matrices

1. Rotation about Z-axis $R_z(\theta)$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2. Translation along Z-axis $T_z(d)$

$$T_z(d) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3. Translation along X-axis $T_x(a)$

$$T_x(a) = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4. Rotation about X-axis $R_x(\alpha)$

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha & 0 \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

🔗 Multiplying the Matrices

We compute:

$${}^A T_B = R_z(\theta) \cdot T_z(d) \cdot T_x(a) \cdot R_x(\alpha)$$

Let's multiply step by step.

Step 1: $A_1 = R_z(\theta) \cdot T_z(d)$

$$A_1 = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 2: $A_2 = A_1 \cdot T_x(a)$

$$A_2 = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & a \cos \theta \\ \sin \theta & \cos \theta & 0 & a \sin \theta \\ 0 & 0 & 1 & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 3: ${}^A T_B = A_2 \cdot R_x(\alpha)$

$${}^A T_B = \begin{bmatrix} \cos \theta & -\sin \theta \cos \alpha & \sin \theta \sin \alpha & a \cos \theta \\ \sin \theta & \cos \theta \cos \alpha & -\cos \theta \sin \alpha & a \sin \theta \\ 0 & \sin \alpha & \cos \alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

✅ Final Answer:

$${}^A T_B = \begin{bmatrix} \cos \theta & -\sin \theta \cos \alpha & \sin \theta \sin \alpha & a \cos \theta \\ \sin \theta & \cos \theta \cos \alpha & -\cos \theta \sin \alpha & a \sin \theta \\ 0 & \sin \alpha & \cos \alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Let me know if you'd like a numerical example with given values for θ , d , a , α .

Given:

jT_i :

$${}^jT_i = \begin{bmatrix} 0.866 & -0.500 & 0 & 11 \\ 0.500 & 0.866 & 0 & -1 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

kT_i :

$${}^kT_i = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.866 & -0.500 & 10 \\ 0 & 0.500 & 0.866 & -20 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

► Goal: Find jT_k

We know:

$${}^jT_k = {}^jT_i \cdot {}^iT_k$$

But we are given kT_i , so we need to compute:

$${}^iT_k = ({}^kT_i)^{-1}$$

Then compute:

$${}^jT_k = {}^jT_i \cdot ({}^kT_i)^{-1}$$

📅 Step 1: Invert kT_i

A homogeneous transformation matrix is of the form:

$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \Rightarrow T^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix}$$

So for kT_i :

Rotation part R :

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.500 \\ 0 & 0.500 & 0.866 \end{bmatrix}$$

$$R^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & 0.500 \\ 0 & -0.500 & 0.866 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -0.500 & 0.866 \end{bmatrix}$$

Position vector p :

$$p = \begin{bmatrix} 0 \\ 10 \\ -20 \end{bmatrix}$$

$$-R^T p = - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & 0.500 \\ 0 & -0.500 & 0.866 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 10 \\ -20 \end{bmatrix} = - \begin{bmatrix} 0 \\ 0.866 \times 10 + 0.5 \times (-20) \\ -0.5 \times 10 + 0.866 \times (-20) \end{bmatrix} = - \begin{bmatrix} 0 \\ 8.66 - 10 = -1.34 \\ -5 - 17.32 = -22.32 \end{bmatrix} = \begin{bmatrix} 0 \\ 1.34 \\ 22.32 \end{bmatrix}$$

So:

$${}^i T_k = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.866 & 0.5 & 1.34 \\ 0 & -0.5 & 0.866 & 22.32 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 2: Multiply ${}^j T_k = {}^j T_i \cdot {}^i T_k$

Let's now compute:

$${}^j T_k = \begin{bmatrix} 0.866 & -0.5 & 0 & 11 \\ 0.5 & 0.866 & 0 & -1 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.866 & 0.5 & 1.34 \\ 0 & -0.5 & 0.866 & 22.32 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix Multiplication Result:

Let's multiply each part:

Rotation (upper-left 3x3):

$$R_{jk} = R_j \cdot R_k = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & 0.5 \\ 0 & -0.5 & 0.866 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.433 & -0.25 \\ 0.5 & 0.75 & 0.433 \\ 0 & -0.5 & 0.866 \end{bmatrix}$$

Translation vector:

$$p_{jk} = R_{jk} \cdot p_k + p_j$$

$$p_k = \begin{bmatrix} 0 \\ 1.34 \\ 22.32 \end{bmatrix}, \quad p_j = \begin{bmatrix} 11 \\ -1 \\ 8 \end{bmatrix}$$

$$R_j \cdot p_k = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1.34 \\ 22.32 \end{bmatrix} = \begin{bmatrix} -0.5 \times 1.34 \\ 0.866 \times 1.34 \\ 22.32 \end{bmatrix} = \begin{bmatrix} -0.67 \\ 1.161 \\ 22.32 \end{bmatrix}$$

$$p_{jk} = \begin{bmatrix} -0.67 \\ 1.161 \\ 22.32 \end{bmatrix} + \begin{bmatrix} 11 \\ -1 \\ 8 \end{bmatrix} = \begin{bmatrix} 10.33 \\ 0.161 \\ 30.32 \end{bmatrix}$$

Final Answer:

$${}^j T_k = \begin{bmatrix} 0.866 & -0.433 & -0.25 & 10.33 \\ 0.5 & 0.75 & 0.433 & 0.161 \\ 0 & -0.5 & 0.866 & 30.32 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Let me know if you'd like it in symbolic form or need help verifying in code.

Statement

Let

$$T = \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix},$$

where R is a 3×3 rotation matrix (so $R^{-1} = R^T$) and $d \in \mathbb{R}^3$. Prove that

$$T^{-1} = \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix}$$

and expand this into component form to match the book's equation (2.105).

1) Algebraic derivation (block-matrix)

Assume the inverse has block form

$$T^{-1} = \begin{bmatrix} S & t \\ 0 & 1 \end{bmatrix},$$

with unknown $S \in \mathbb{R}^{3 \times 3}$ and $t \in \mathbb{R}^3$. By definition,

$$T^{-1}T = \begin{bmatrix} S & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} SR & Sd + t \\ 0 & 1 \end{bmatrix}.$$

This must equal the 4×4 identity $\begin{bmatrix} I_3 & 0 \\ 0 & 1 \end{bmatrix}$. Hence

$$SR = I_3, \quad Sd + t = 0.$$

From $SR = I_3$ we get $S = R^{-1} = R^T$ (because R is orthonormal). From $Sd + t = 0$ we get $t = -Sd = -R^T d$. Substituting,

$$T^{-1} = \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix}.$$

2) Component expansion

Write $R = [n \ o \ a]$ (columns n, o, a) and $d = [d_x \ d_y \ d_z]^T$. Then

$$R^T = \begin{bmatrix} n_x & n_y & n_z \\ o_x & o_y & o_z \\ a_x & a_y & a_z \end{bmatrix}, \quad -R^T d = \begin{bmatrix} -d \cdot n \\ -d \cdot o \\ -d \cdot a \end{bmatrix}.$$

So

$$T^{-1} = \begin{bmatrix} n_x & n_y & n_z & -d \cdot n \\ o_x & o_y & o_z & -d \cdot o \\ a_x & a_y & a_z & -d \cdot a \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

which is exactly equation (2.105).

3) Direct verification

Multiply the proposed inverse by T :

$$\begin{aligned} \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix} &= \begin{bmatrix} R^T R & R^T d - R^T d \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} I_3 & 0 \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

So the formula is correct.

3) Direct verification

Multiply the proposed inverse by T :

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So the formula is correct.

4) Geometric interpretation (why $-R^T d$)

- T transforms a point p expressed in the local (body) frame to the world frame by $p_w = R p_b + d$.
- To invert, given p_w we recover the local coordinates by first **subtracting the translation** (move origin back) and then **undoing the rotation**. However, because rotation and translation do not commute, the order and rotation matter:

$$p_b = R^{-1}(p_w - d) = R^T p_w - R^T d.$$

- So the new origin of the world frame expressed in the body frame is $-R^T d$: that is, from the body frame's point of view the world origin sits at $-R^T d$.
-

5) Numeric example

Take a simple rotation of 90° about the z -axis:

$$R = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad d = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Compute R^T :

$$R^T = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Compute $R^T d$ (step-by-step):

- first component: $0 \cdot 1 + 1 \cdot 2 + 0 \cdot 3 = 2$,
- second: $-1 \cdot 1 + 0 \cdot 2 + 0 \cdot 3 = -1$,
- third: $0 \cdot 1 + 0 \cdot 2 + 1 \cdot 3 = 3$.

So $R^T d = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ and

$$-R^T d = \begin{bmatrix} -2 \\ 1 \\ -3 \end{bmatrix}.$$

Thus

$$T^{-1} = \begin{bmatrix} 0 & 1 & 0 & -2 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Multiplying $T^{-1}T$ yields the identity (because $R^T R = I_3$ and $R^T d - R^T d = 0$), confirming the result numerically.

Final boxed result

$$\boxed{\begin{array}{l} T = \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix} \implies T^{-1} = \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix}, \\ \text{equivalently } T^{-1} = \begin{bmatrix} n_x & n_y & n_z & -d \cdot n \\ o_x & o_y & o_z & -d \cdot o \\ a_x & a_y & a_z & -d \cdot a \\ 0 & 0 & 0 & 1 \end{bmatrix}. \end{array}}$$
